

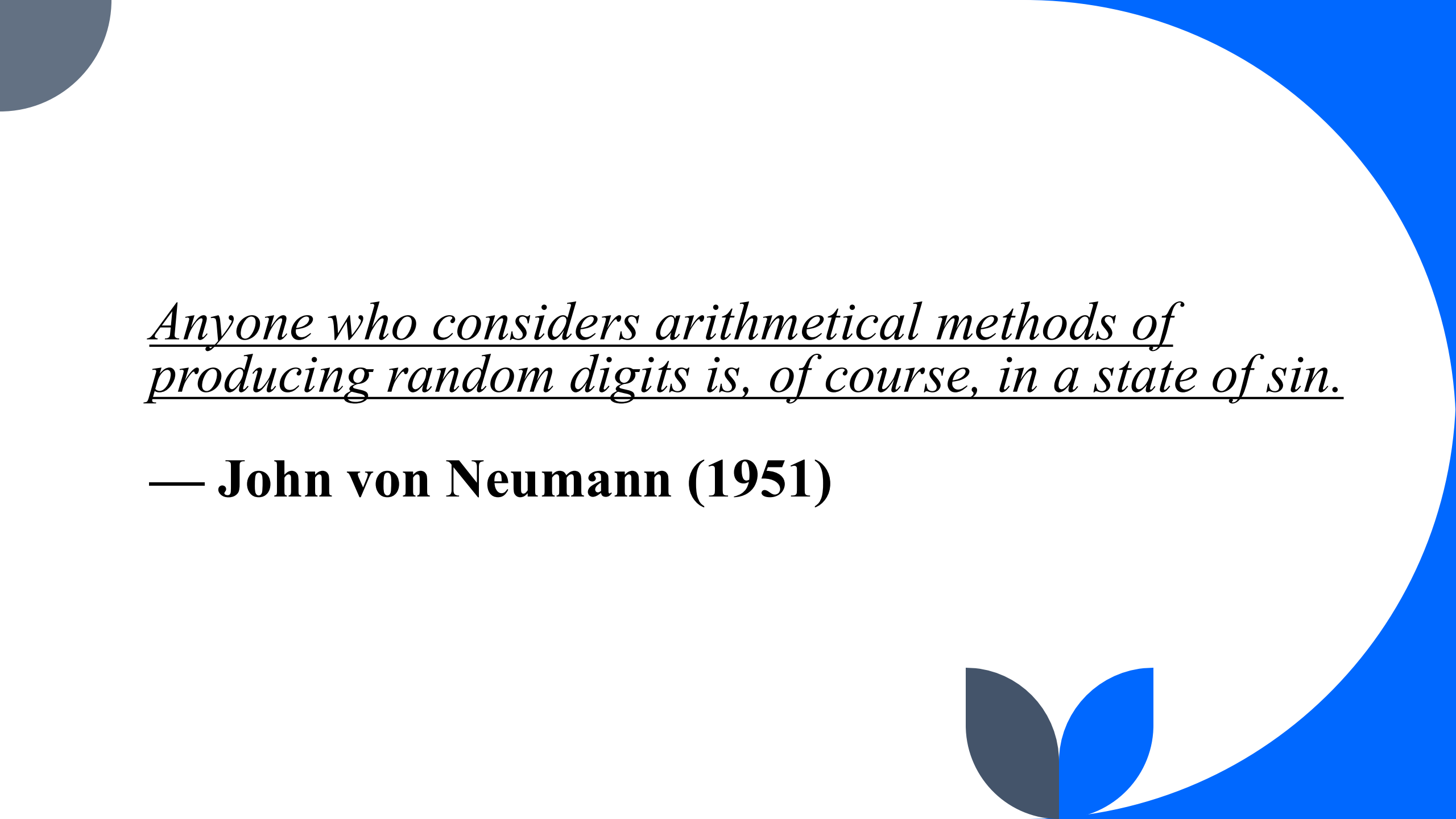
EntropyX

*A Physical True Random Number
Generator*

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*Anyone who considers arithmetical methods of
producing random digits is, of course, in a state of sin.*

— **John von Neumann (1951)**



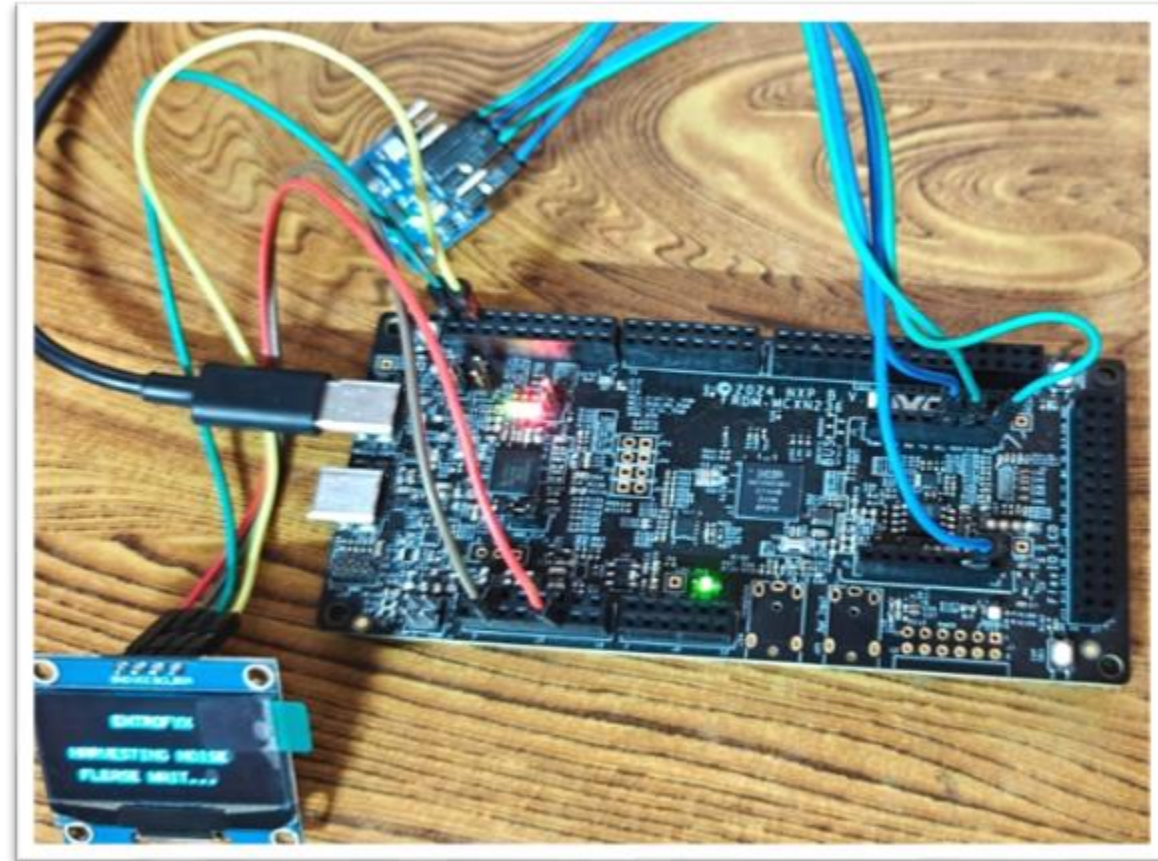
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INTRODUCTION

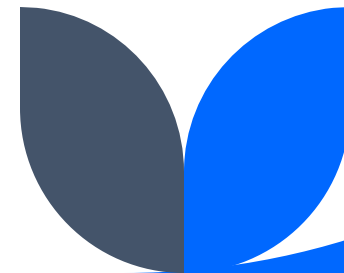
EntropyX is a physical **True Random Number Generator (TRNG)** and key-derivation device.

It harvests physical noise across three independent channels **to generate random number using algorithms which is unpredictable .**



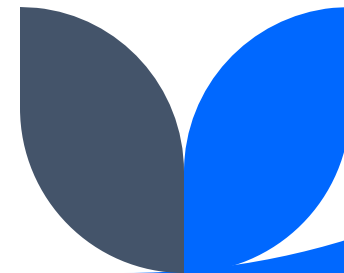
PROBLEM STATEMENT

- **Software-based Pseudo-Random Number Generators (PRNGs)** rely on predictable algorithms.
- When initialized with predictable system parameters such as clock "**timestamps**" or "**memory buffers**", the entire sequence of cryptographic keys can be computed by an adversary.

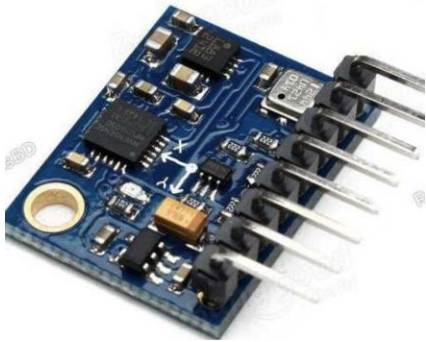


OBJECTIVES

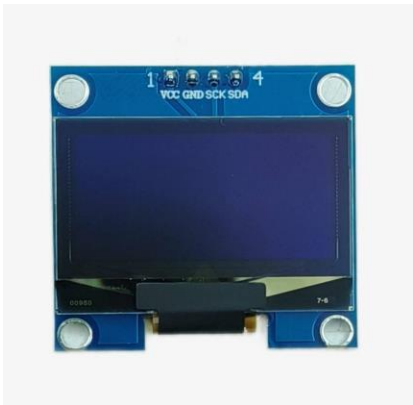
- **Creating True Random Number Generator (TRNG)** using the cost effective hardware under ₹ 3800.
- **The Today's Commercial Hardware Security Modules(HSMs)** address this using physical noise sources, but they cost between ₹45,000 and ₹2,00,000, function as closed-source black boxes, and do not expose internal entropy metrics in real time.



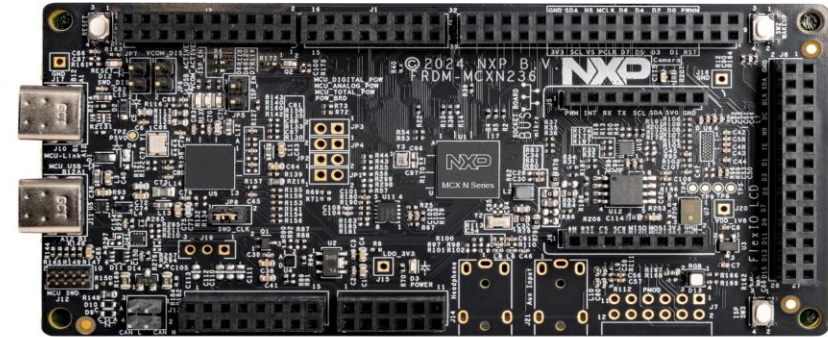
ELECTRONIC COMPONENTS USED



**GY-87 with
MPU6050 10DOF Sensor
Module MPU6050
HMC5883L BMP180**



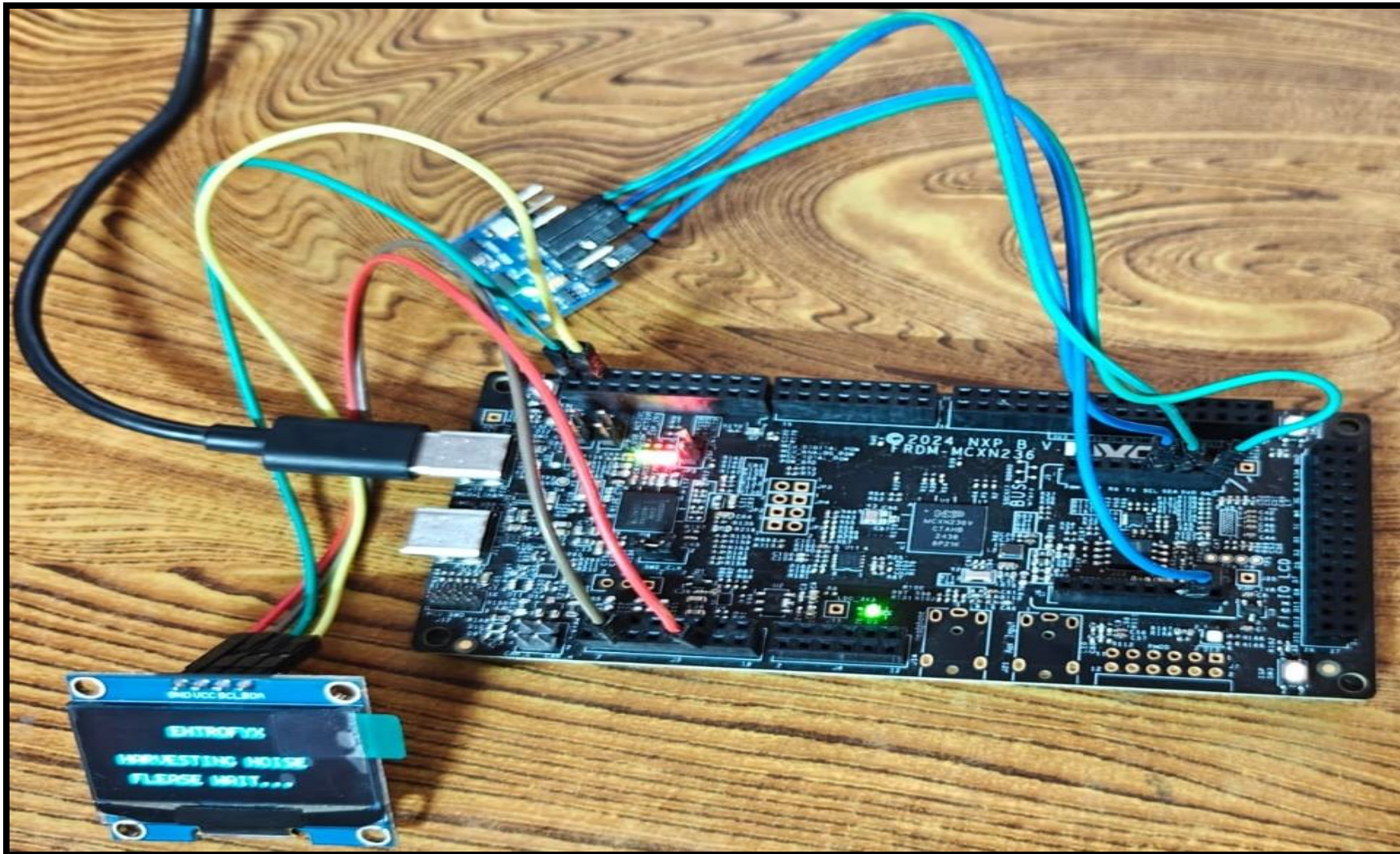
**1.3 INCH 12C/IIC OLED
DISPLAY MODULE -
SSH1106 CONTROLLER -**



**FRDM-MCXN236 –MICRO CONTROLLER
BOARD –POWERED BY AN ARM-CORTEX -M33**



PROJECT CONNECTED AND IN WORKING CONDITION



HOW IT WORKS?

- Instead of using software formulas, EntropyX listens to the microscopic noise from the sensors on board- magnetic field, acceleration, gyroscope,
- Then a built in digital filters all the static forces such as gravity and after which the microcontroller tests the noise(the entropy) with the official NIST standards , rejecting all the predictable data with strict security features while presenting only the SHA256 Military Grade uncrackable 256-bit key.



The Math Behind it!

- **Removing Gravity:**

$$\Delta S(t) = S(t) - S(t-1)$$

- **Von Neumann De-biasing**

$$P(0,1)=P(1,0)=p(1-p)$$

If a physical sensor produces slightly more 1s than 0s, we flip it twice!
"Heads then Tails" has the exact same probability as "Tails then Heads."
This guarantees an exact 50/50 balance.



- **Measuring True Randomness (NIST SP 800-90B Min-Entropy)**

$$H_{\infty} = -\log_2 (p_{\max})$$

Measures the chance of an attacker guessing the key on their first try. If the noise isn't unpredictable enough, the chip rejects it.

- **Leftover Hash Lemma**

Uses a Toeplitz universal hash matrix to convert 1,024 raw physical bits into 256 pure, uniform key bits, that will mathematically be impossible to reverse engineer.

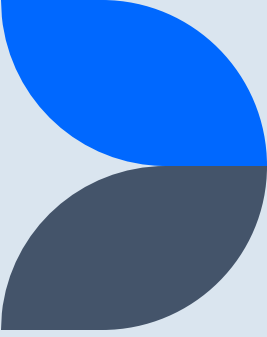


Web Dashboard – For Verifying Live Data



CONCLUSION

- EntropyX shows that secure random number can be generated using natural physical sources of randomness. It combines different source of tiny unpredictable changes to create strong random numbers, without needing the internet .
- My project is low cost, compact, and designed to provide a more secure way of generating keys and protecting digital information.





Thank you

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